One end of the leaf spring is attached to the frame through shackle as movable. Another end is attached with anchor pin as non movable.

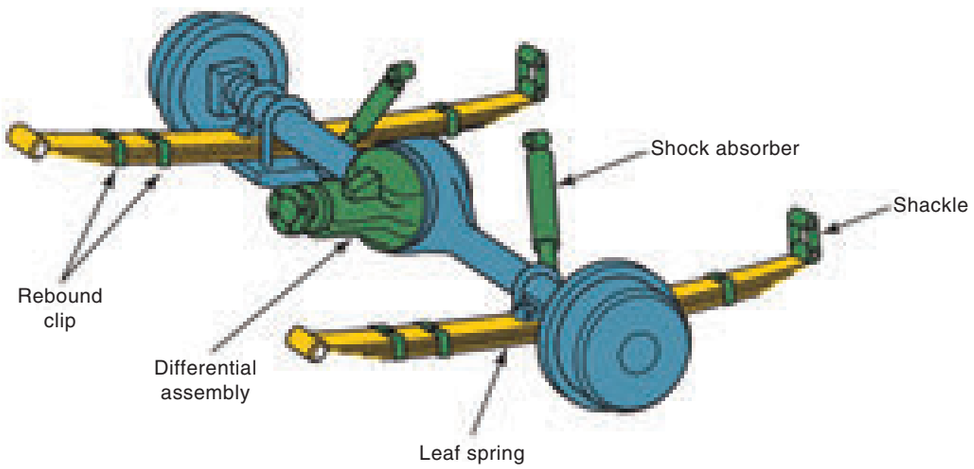
Working Principle

When the vehicle travel over a road undulation, the force from the road act on the leaf spring and the spring gets straightened. This variation in length is accommodated by the oscillation movement of the shackle. Thus, the road force makes the deflection of the leaf spring and absorb the vibration. During deflection, the energy is stored in the leaf spring and the stored energy is dissipated when the leaf spring retains its original position. Thus the road shocks are reduced and the ride comfort is increased.

In light commercial vehicles spring rubber or hard plastic bush are used. Phosphorus bronze bushes are used in heavy commercial vehicles. The lubricating oil with graphite is used in the bushes for lubrication.

Material

Chromium vanadium steel, Chromium- Nickel- Molybdenum steel, Silicon- manganese steel, are the typical materials that are used in the manufacturing of leaf springs.



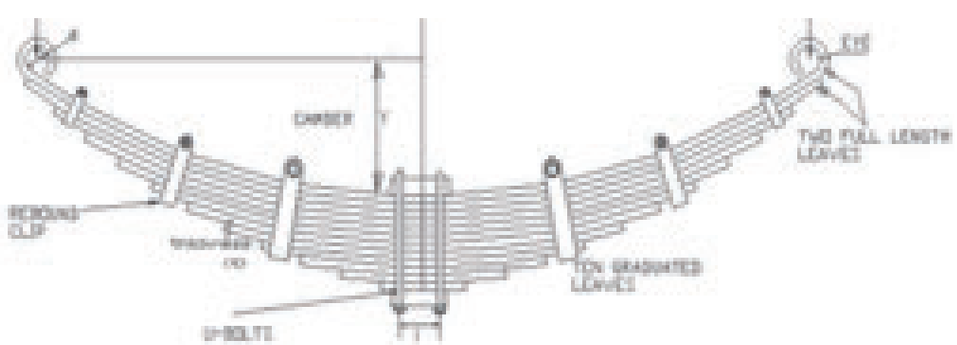


Figure 7.1.1 Leaf Spring

BAE_ENG_Chapter_07.indd 125 12/10/2019 11:54:02 AM